

Calculus 2

6 Lecture 6

Properties of Limit Functions

Definition 6.1. Let $E \subseteq \mathbb{R}$. Then, $\lambda \in \mathbb{R}$ is called a *limit point* (or *accumulation point*) of a set E if

$$\forall \varepsilon > 0 \quad \exists x \in E \setminus \{\lambda\} \quad \rightarrow \quad |\lambda - x| < \varepsilon.$$

Claim 6.1 (Theorem on Changing the Order of Two Limits). Let $E \subseteq \mathbb{R}$ and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) λ is a *finite limit point* of E (finite means that $\lambda \neq \pm\infty$);
- b) $\forall n \in \mathbb{N} \rightarrow \exists \lim_{x \rightarrow \lambda} f_n(x) = \alpha_n$;
- c) $f_n \xrightarrow{E} f$ (see [Def 5.3](#)),

then:

1. $\exists \lim_{n \rightarrow \infty} \alpha_n$, that is, the sequence $\{\alpha_n\}_{n=1}^{\infty}$ converges;
- 2.

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow \lambda} f_n(x) \right) = \lim_{x \rightarrow \lambda} \left(\lim_{n \rightarrow \infty} f_n(x) \right). \quad (1)$$

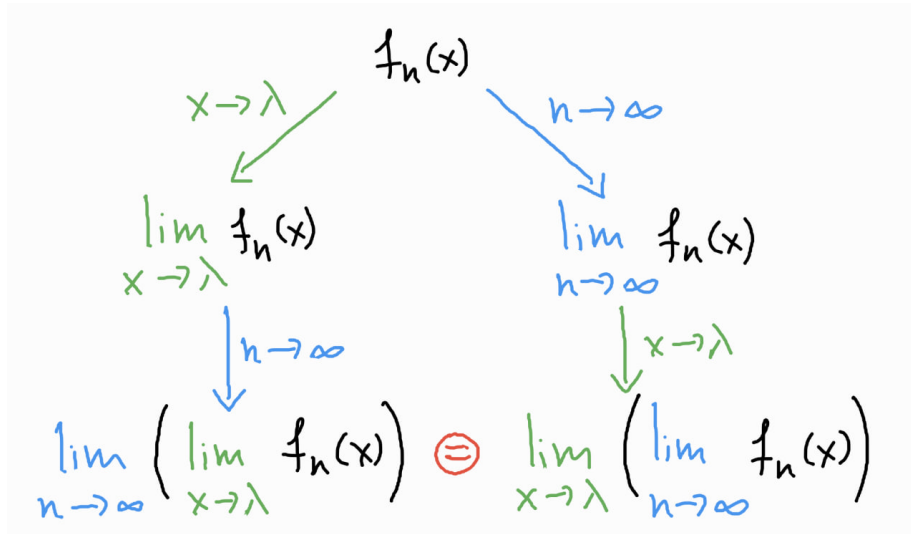


Figure 6.1

The equality in [Formula \(1\)](#) is the statement of [Item 2](#) of [Claim 6.1](#) and the reason for the name of the claim.

Proof. 1) Since, due to [Item b](#)), $\forall n \in \mathbb{N} \rightarrow \exists \lim_{x \rightarrow \lambda} f_n(x) = \alpha_n$, we have

$$(\forall x \rightarrow \lambda \quad |f_m(x) - f_k(x)| < \varepsilon) \implies (|\alpha_m - \alpha_k| \leq \varepsilon). \quad (*)$$

Now note that:

$$\begin{aligned} & (f_n \xrightarrow{E} f) \\ \Updownarrow & \text{ due to } \text{Claim 5.4} \end{aligned}$$

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \forall x \in E \rightarrow |f_m(x) - f_k(x)| < \varepsilon)$$

$\Downarrow$ due to [Formula \(*\)](#)

$$(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall k \geq N(\varepsilon) \forall m \geq N(\varepsilon) \rightarrow |\alpha_m - \alpha_k| \leq \varepsilon)$$

$\Updownarrow$ due to the Cauchy criterion for convergence of numerical sequences (see Calculus-1)

$$\left(\exists \lim_{n \rightarrow \infty} \alpha_n \right).$$

2) Let

$$\lim_{n \rightarrow \infty} \alpha_n = \alpha \quad (\text{the limit exists due to [Item 1](#)}).$$

Then, in [Formula \(1\)](#):

the left hand side is

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow \lambda} f_n(x) \right) = \lim_{n \rightarrow \infty} \alpha_n = \alpha,$$

the right hand side is

$$\lim_{x \rightarrow \lambda} \left(\lim_{n \rightarrow \infty} f_n(x) \right) = \lim_{x \rightarrow \lambda} f(x),$$

since, due to [Item c](#)) and [Claim 5.3](#), we have $f_n \xrightarrow{E} f$.

Thus, we need to prove that

$$\exists \lim_{x \rightarrow \lambda} f(x) = \alpha.$$

That is, due to a definition of the limit (see Calculus-1), we need to prove that

$$\forall \varepsilon > 0 \exists \delta > 0 : \forall x \in (\alpha - \delta, \alpha + \delta) \cap E \rightarrow |f(x) - \alpha| < \varepsilon. \quad (**)$$

Note that, for some $\varepsilon > 0$ we have:

i)

$$(f_n \xrightarrow{E} f) \implies \left(\exists N_1(\varepsilon) : \forall n \geq N_1(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{3} \right). \quad (1a)$$

ii)

$$\left(\lim_{n \rightarrow \infty} \alpha_n = \alpha \right) \implies \left(\exists N_2 : \forall n \geq N_2 \rightarrow |\alpha_n - \alpha| < \frac{\varepsilon}{3} \right). \quad (2)$$

Let $N(\varepsilon) = \max\{N_1(\varepsilon), N_2\}$. Then

$$\left(\lim_{x \rightarrow \lambda} f_N(x) = \alpha_N \right)$$

(for simplicity of notation we write N instead of $N(\varepsilon)$), hence

$$\left(\exists \delta > 0 : \forall x \in (\alpha - \delta, \alpha + \delta) \cap E \rightarrow |f_N(x) - \alpha_N| < \frac{\varepsilon}{3} \right). \quad (3)$$

Now we are ready to verify (**). Indeed, for every $x \in (\alpha - \delta, \alpha + \delta)$ we have

$$\begin{aligned} |f(x) - \alpha| &= |f(x) - f_N(x) + f_N(x) - \alpha_N + \alpha_N - \alpha| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - \alpha_N| + |\alpha_N - \alpha| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon, \end{aligned}$$

due to (1a), (3), and (2), respectively.

Thus, we verified (**) and proved [Item 2](#). □

Claim 6.2 (Continuity of a Limit Function). Let $E \subseteq \mathbb{R}$ and $f_n : E \rightarrow \mathbb{R}, \forall n \in \mathbb{N}$. If:

a) for every $n \in \mathbb{N}$, f_n is a continuous function on E , that is, $\forall n \in \mathbb{N} \rightarrow f_n \in C(E)$;

b) $f_n \xrightarrow{E} f$ (see [Def 5.3](#)),

then, the function f is continuous on E . That is, $f \in C(E)$.

Compare [Claim 6.2](#) with [Item 1](#)) of [Claim 5.1](#) and [Claim 5.2](#).

Proof. If $\lambda \in E$ is an *isolated point* of E (that is, if λ is an element of E and there exists a neighborhood of λ which does not contain any other elements of E), then, f is continuous at λ (see Calculus-1).

If $\lambda \in E$ is a *limit point* of E , then from Calculus-1, it is known that

$$(f \text{ is continuous at } \lambda \in E) \iff \left(\lim_{x \rightarrow \lambda} f(x) = f(\lambda) \right). \quad (4)$$

Therefore, it remains to prove that for every limit point $\lambda \in E$ we have $\lim_{x \rightarrow \lambda} f(x) = f(\lambda)$.

Let $\lambda \in E$ be a limit point of E . Then, due to [Item a\)](#) and [Expression \(4\)](#), we have

$$\forall n \in \mathbb{N} \rightarrow \lim_{x \rightarrow \lambda} f_n(x) = f_n(\lambda) = \alpha_n.$$

Therefore, all conditions of [Claim 6.1](#) are satisfied and we have

$$\lim_{x \rightarrow \lambda} f(x) = \lim_{x \rightarrow \lambda} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow \lambda} f_n(x) = \lim_{n \rightarrow \infty} f_n(\lambda) = f(\lambda),$$

since $(f_n \xrightarrow{E} f) \Rightarrow (f_n \xrightarrow{E} f)$, due to [Item 2\)](#) of [Claim 6.1](#), since $\forall n \in \mathbb{N}$ f_n is continuous on E , and since $(f_n \xrightarrow{E} f) \Rightarrow (f_n \xrightarrow{E} f)$.

That is, the claim is proved. □

To prove the next claim, we will need the following fact.

Claim 6.3 (The Darboux Criterion for Riemann Integrability). Let $I = [a, b] \subseteq \mathbb{R}$ and $f : I \rightarrow \mathbb{R}$. Then

$$(f \text{ is Riemann integrable on } I = [a, b], \text{ notation } f \in \mathcal{R}([a, b]))$$

$$\Updownarrow$$

$$\left(\forall \varepsilon > 0 \exists d(\varepsilon) > 0 \forall T : |T| < d(\varepsilon) \rightarrow \sum_{I_i \in T} \omega_{I_i}(f) |I_i| < \varepsilon \right). \quad (5)$$

where: T is a *partition* of $I = [a, b]$ (see Calculus-1); I_i is a *subinterval* of I from the partition T ; $|I_i|$ is the *length* of a subinterval I_i (if $I_i = [x_{i-1}, x_i]$, then $|I_i| = x_i - x_{i-1}$);

$$|T| = \max_{I_i \in T} |I_i|$$

is the *mesh* of the partition T ; and

$$\omega_{I_i}(f) = \sup_{x \in I_i} f(x) - \inf_{x \in I_i} f(x)$$

is the *oscillation* of f on an interval I_i .

Proof. See Calculus-1. □

Claim 6.4 (Riemann Integrability of a Limit Function). Let $I = [a, b] \subset \mathbb{R}$ and $f_n : I \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

a) for every $n \in \mathbb{N}$, f_n is Riemann integrable on $I = [a, b]$, that is, $\forall n \in \mathbb{N} \rightarrow f_n \in \mathcal{R}([a, b])$;

b) $f_n \xrightarrow{I} f$ (see [Def 5.3](#)),

then, the function f is Riemann integrable on I , that is, $f \in \mathcal{R}([a, b])$.

Compare [Claim 6.4](#) with [Item 3](#)) of [Claim 5.1](#) and [Claim 5.2](#).

Proof. Due to [Claim 6.3](#) and [Formula \(5\)](#) it suffices to show that

$$\forall \varepsilon > 0 \exists d(\varepsilon) > 0 \forall T : |T| < d(\varepsilon) \rightarrow \sum_{I_i \in T} \omega_{I_i}(f) |I_i| < \varepsilon. \quad (**)$$

For some $\varepsilon > 0$, note that:

i)

$$(f_n \xrightarrow{I} f) \implies \left(\exists N(\varepsilon) \forall x \in I \rightarrow |f(x) - f_N(x)| < \frac{\varepsilon}{4|I|} \right). \quad (6)$$

For simplicity of notation we write N instead of $N(\varepsilon)$, and $|I| = b - a$.

If we know that

$$\forall x \in I \rightarrow |f(x) - f_N(x)| < \frac{\varepsilon}{4|I|},$$

then

$$\forall x \in I \rightarrow f_N(x) - \frac{\varepsilon}{4|I|} < f(x) < f_N(x) + \frac{\varepsilon}{4|I|}.$$

Then, for every subset $J \subseteq I$, we have

$$\inf_{x \in J} f_N(x) - \frac{\varepsilon}{4|I|} \leq \inf_{x \in J} f(x) \quad (7a)$$

and

$$\sup_{x \in J} f(x) \leq \sup_{x \in J} f_N(x) + \frac{\varepsilon}{4|I|}. \quad (7b)$$

Therefore, due to [\(6\)](#), for every subset $J \subseteq I$, we have

$$\begin{aligned} \omega_J(f) &= \sup_{x \in J} f(x) - \inf_{x \in J} f(x) \\ &\leq \sup_{x \in J} f_N(x) + \frac{\varepsilon}{4|I|} - \left(\inf_{x \in J} f_N(x) - \frac{\varepsilon}{4|I|} \right) \\ &= \omega_J(f_N) + \frac{\varepsilon}{2|I|}. \end{aligned} \quad (8)$$

ii) For f_N from [Formula \(6\)](#), due to [Claim 6.3](#), we have

$$(f_N \in \mathcal{R}(I)) \implies \left(\exists d(\varepsilon) > 0 \forall T : |T| < d(\varepsilon) \rightarrow \sum_{I_i \in T} \omega_{I_i}(f_N) |I_i| < \frac{\varepsilon}{2} \right). \quad (9)$$

This holds true due to [Item a\)](#). Here, for a given $\varepsilon > 0$ and $N(\varepsilon)$, we define $d(\varepsilon)$ which we are going to use to verify [\(**\)](#).

We will show now that $d(\varepsilon)$ from [Formula \(9\)](#) satisfies [\(**\)](#). For every partition T of I such that $|T| < d(\varepsilon)$, we have

$$\begin{aligned} \sum_{I_i \in T} \omega_{I_i}(f) |I_i| &\leq \sum_{I_i \in T} \left(\omega_{I_i}(f_N) + \frac{\varepsilon}{2|I|} \right) |I_i| \\ &= \sum_{I_i \in T} \omega_{I_i}(f_N) |I_i| + \frac{\varepsilon}{2|I|} \sum_{I_i \in T} |I_i| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2|I|} |I| = \varepsilon. \end{aligned}$$

The first inequality holds due to [Inequality \(8\)](#), which holds true for every subset J of I ; the second term uses $\sum_{I_i \in T} |I_i| = |I|$, since we sum up the lengths of all subintervals from the partition T .

Thus, we verified [\(**\)](#) and proved the claim. $\square$

Claim 6.5 (Integration of a Uniformly Convergent Sequence). Let $I = [a, b] \subset \mathbb{R}$ and $f_n : I \rightarrow \mathbb{R}$. If:

- a) for every $n \in \mathbb{N}$, f_n is Riemann integrable on I , that is, $\forall n \in \mathbb{N} \rightarrow f_n \in \mathcal{R}(I)$;
- b) $f_n \xrightarrow{I} f$,

then

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx.$$

Proof. Due to [Claim 6.4](#), the limit function

$$f(x) = \lim_{n \rightarrow \infty} f_n(x), \quad x \in I,$$

is Riemann integrable on I . Therefore, exists the integral:

$$\int_a^b f(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx.$$

Due to [Item b\)](#) and [Def 5.3](#), we have:

$$(f_n \xrightarrow{I} f) \implies \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} \forall n \geq N(\varepsilon) \forall x \in I \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{b-a} \right). \quad (10)$$

Therefore, for $\forall \varepsilon > 0$ and $\forall n \geq N(\varepsilon)$ we have

$$\begin{aligned} \left| \int_a^b f_n(x) dx - \int_a^b f(x) dx \right| &= \left| \int_a^b (f_n(x) - f(x)) dx \right| \\ &\leq \int_a^b |f_n(x) - f(x)| dx \\ &\leq \int_a^b \frac{\varepsilon}{b-a} dx = \frac{\varepsilon}{b-a} (b-a) = \varepsilon. \end{aligned}$$

The first equality uses linearity of the Riemann integral, and the inequality

$$\left| \int_a^b g(x) dx \right| \leq \int_a^b |g(x)| dx, \quad \forall g \in \mathcal{R}([a, b]),$$

see Calculus-1.

Thus, due to a definition of the limit of a numerical sequence (see Calculus-1), we proved that

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx.$$

$\square$

Note that if all the functions f_n are differentiable on an interval $J = (a, b)$ and converge uniformly to a limit function f on J , the limit function f need not be differentiable.

Example. Let $J = (0, 2\pi)$ and

$$f_n(x) = \sum_{k=1}^n \frac{\sin(k^2 x)}{k^2}, \quad \forall n \in \mathbb{N}.$$

Then, due to [Claim 5.7](#) [The Weierstrass M-test], we have

$$f_n \xrightarrow{J} f, \quad \text{where } f(x) = \sum_{k=1}^{\infty} \frac{\sin(k^2 x)}{k^2}$$

is called the Riemann function.

It is clear that the functions f_n are differentiable on J . It can be shown (though it is rather difficult and is therefore not included in our course) that the Riemann function f is differentiable almost nowhere on J .

That is, a direct analogue of [Claim 6.2](#) and [Claim 6.4](#) does not hold true for differentiation. Our next goal is to establish conditions under which the limit function of a sequence of differentiable functions is a differentiable function.

Claim 6.6. Let $E \subseteq \mathbb{R}$, $A \subseteq E$, $B \subseteq E$, and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) $f_n \xrightarrow{E} f$ (see [Def 5.2](#));
- b) $f_n \xrightarrow{A} f$ and $f_n \xrightarrow{B} f$ (see [Def 5.3](#)),

then

$$f_n \xrightarrow{A \cup B} f.$$

Proof. Follows directly from [Def 5.3](#):

$$(f_n \xrightarrow{A} f) \iff (\forall \varepsilon > 0 \exists N_1(\varepsilon) \forall n \geq N_1(\varepsilon) \forall x \in A \rightarrow |f_n(x) - f(x)| < \varepsilon), \quad (11a)$$

$$(f_n \xrightarrow{B} f) \iff (\forall \varepsilon > 0 \exists N_2(\varepsilon) \forall n \geq N_2(\varepsilon) \forall x \in B \rightarrow |f_n(x) - f(x)| < \varepsilon). \quad (11b)$$

Let $N(\varepsilon) = \max\{N_1, N_2\}$. Then, from Formulas (11a)–(11b), we have:

$$\forall \varepsilon > 0 \exists N(\varepsilon) \forall n \geq N(\varepsilon) \forall x \in A \cup B \rightarrow |f_n(x) - f(x)| < \varepsilon.$$

Therefore, due to [Def 5.3](#), we have $f_n \xrightarrow{A \cup B} f$. □

Claim 6.7. Let $E \subseteq \mathbb{R}$, $g : E \rightarrow \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) $f_n \xrightarrow{E} f$;
- b) g is bounded on E (that is, $\exists M > 0 \forall x \in E \rightarrow |g(x)| \leq M$),

then

$$g \cdot f_n \xrightarrow{E} g \cdot f.$$

That is, when multiplying a uniformly convergent sequence by a bounded function, we obtain a uniformly convergent sequence.

Proof. Due to [Item b\)](#), we have $\exists M > 0 \forall x \in E \rightarrow |g(x)| \leq M$. Then, due to [Def 5.3](#), we have:

$$(f_n \xrightarrow{E} f) \implies \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{M} \right). \quad (12)$$

Since, for every $x \in E$, we have

$$|g(x)f_n(x) - g(x)f(x)| = |g(x)| |f_n(x) - f(x)| \leq M |f_n(x) - f(x)|,$$

[Formula \(12\)](#) implies that

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} \forall n \geq N(\varepsilon) \forall x \in E \rightarrow |g(x)f_n(x) - g(x)f(x)| < \varepsilon,$$

which, due to [Def 5.3](#), means that $g f_n \xrightarrow{E} g f$. □

Claim 6.8. Let $E \subseteq \mathbb{R}$, $f_n : E \rightarrow \mathbb{R}$, and $g_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If $f_n \xrightarrow{E} f$ and $g_n \xrightarrow{E} g$, then, $f_n + g_n \xrightarrow{E} f + g$.

Proof. Follows directly from [Def 5.3](#):

$$(f_n \xrightarrow{E} f) \implies \left(\forall \varepsilon > 0 \exists N_1(\varepsilon) \forall n \geq N_1(\varepsilon) \forall x \in E \rightarrow |f_n(x) - f(x)| < \frac{\varepsilon}{2} \right), \quad (12a)$$

$$(g_n \xrightarrow{E} g) \implies \left(\forall \varepsilon > 0 \exists N_2(\varepsilon) \forall n \geq N_2(\varepsilon) \forall x \in E \rightarrow |g_n(x) - g(x)| < \frac{\varepsilon}{2} \right). \quad (12b)$$

Let $N(\varepsilon) = \max\{N_1(\varepsilon), N_2(\varepsilon)\}$. Then, from Formulas (12a)–(12b), $\forall \varepsilon > 0$, $\forall n \geq N(\varepsilon)$, $\forall x \in E$ we have

$$|f_n(x) + g_n(x) - f(x) - g(x)| \leq |f_n(x) - f(x)| + |g_n(x) - g(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which, due to [Def 5.3](#), implies that $f_n + g_n \xrightarrow{E} f + g$. □

Claim 6.9. Let $E \subseteq \mathbb{R}$, and $f_n : E \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If E is a *finite set*, then

$$(f_n \xrightarrow{E} f) \iff (f_n \xrightarrow{E} f).$$

Proof. ($\Rightarrow$) Let $E = \{\alpha_1, \alpha_2, \dots, \alpha_K\}$. Then, due to [Def 5.2](#), for every $\alpha_k \in E$, we have

$$(f_n \xrightarrow{E} f) \implies (\forall \varepsilon > 0 \exists N(\varepsilon, \alpha_k) \in \mathbb{N} \forall n \geq N(\varepsilon, \alpha_k) \rightarrow |f_n(\alpha_k) - f(\alpha_k)| < \varepsilon). \quad (13)$$

Let

$$N(\varepsilon) = \max\{N(\varepsilon, \alpha_1), N(\varepsilon, \alpha_2), \dots, N(\varepsilon, \alpha_K)\}.$$

Then, from [Formula \(13\)](#), $\forall \varepsilon > 0$, $\forall n \geq N(\varepsilon)$, $\forall x \in E$, we have

$$|f_n(x) - f(x)| < \varepsilon,$$

which, due to [Def 5.3](#), implies that $f_n \xrightarrow{E} f$.

($\Leftarrow$) The claim holds true for every set E (not necessarily finite) due to [Claim 5.3](#). □

Claim 6.10 (Differentiability of a Limit Function). Let $J = (a, b) \subset \mathbb{R}$ and $f_n : J \rightarrow \mathbb{R}$, $\forall n \in \mathbb{N}$. If:

- a) for every $n \in \mathbb{N}$, f_n is differentiable on $J = (a, b)$, that is, $\forall n \in \mathbb{N} \forall x \in (a, b) \rightarrow \exists f'_n(x) \stackrel{\text{def}}{=} \lim_{\delta \rightarrow 0} \frac{f_n(x + \delta) - f_n(x)}{\delta}$;
- b) there is $\lambda \in J = (a, b)$ such that $\exists \lim_{n \rightarrow \infty} f_n(\lambda)$;
- c) the sequence $\{f'_n(x)\}_{n=1}^{\infty}$ converges uniformly on $J = (a, b)$, that is, $f'_n \xrightarrow{J} g$ for some function $g : J \rightarrow \mathbb{R}$,

then:

- 1. $f_n \xrightarrow{J} f$;
- 2. the limit function f is differentiable on J and

$$\forall x \in J \rightarrow f'(x) = \lim_{n \rightarrow \infty} f'_n(x).$$

On the diagram, the green shows what is given and the red shows what need to be proved.

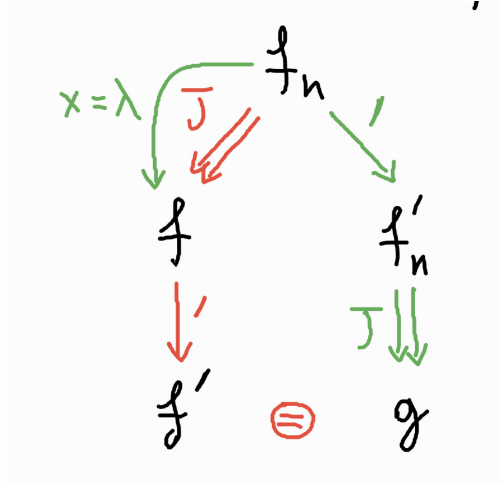


Figure 6.2

Proof. 1) Note that, due to [Item c\)](#) and [Claim 5.4](#), for every $\varepsilon > 0$ there is $N(\varepsilon) \in \mathbb{N}$ such that

$$\forall k \geq N(\varepsilon) \quad \forall m \geq N(\varepsilon) \quad \forall x \in J \rightarrow |f'_k(x) - f'_m(x)| < \varepsilon. \quad (14)$$

For any $y \in J$, let

$$J_y = J \setminus \{y\} = (a, y) \cup (y, b).$$

Consider the functions

$$h_n : J_y \rightarrow \mathbb{R}, \quad h_n(x) = \frac{f_n(x) - f_n(y)}{x - y}, \quad \forall n \in \mathbb{N}, \quad \forall x \in J_y. \quad (***)$$

Let us prove that the sequence

$$\left\{ h_n(x) = \frac{f_n(x) - f_n(y)}{x - y} \right\}_{n=1}^{\infty}$$

converges uniformly on J_y .

Due to [Claim 5.4](#), it suffices to show that

$$\forall \varepsilon > 0 \quad \exists N(\varepsilon) \in \mathbb{N} \quad \forall k \geq N(\varepsilon) \quad \forall m \geq N(\varepsilon) \quad \forall x \in J_y \rightarrow |h_k(x) - h_m(x)| < \varepsilon. \quad (15)$$

Let $N(\varepsilon) \in \mathbb{N}$ be defined as in [\(14\)](#). Then, $\forall k \geq N(\varepsilon)$, $\forall m \geq N(\varepsilon)$, $\forall x \in J_y$ we have

$$\begin{aligned} |h_k(x) - h_m(x)| &= \left| \frac{f_k(x) - f_k(y)}{x - y} - \frac{f_m(x) - f_m(y)}{x - y} \right| \\ &= \left| \frac{(f_k(x) - f_m(x)) - (f_k(y) - f_m(y))}{x - y} \right|. \end{aligned} \quad (16)$$

Note that, due to [Item a\)](#), the function $f_k - f_m$ is continuous on the closed interval

$$[\min\{x, y\}, \max\{x, y\}]$$

and differentiable on the open interval

$$(\min\{x, y\}, \max\{x, y\}).$$

Therefore, due to the Mean Value Theorem (see Calculus-1), we have

$$(f_k(x) - f_m(x)) - (f_k(y) - f_m(y)) = (f'_k(z) - f'_m(z))(x - y),$$

for some $z \in (\min\{x, y\}, \max\{x, y\})$. Thus, due to [Equality \(16\)](#), we have

$$|h_k(x) - h_m(x)| = \left| \frac{(f'_k(z) - f'_m(z))(x - y)}{x - y} \right| = |f'_k(z) - f'_m(z)| < \varepsilon,$$

for some $z \in (\min\{x, y\}, \max\{x, y\})$, since [\(14\)](#) holds true for every $x \in J$ and

$$z \in (\min\{x, y\}, \max\{x, y\}) \subset J.$$

That is, we verified [\(15\)](#). Therefore the sequence

$$\left\{ h_n(x) = \frac{f_n(x) - f_n(y)}{x - y} \right\}_{n=1}^{\infty} \text{ converges uniformly on } J_y = J \setminus \{y\}. \quad (17)$$

Let in [Formula \(17\)](#) $y = \lambda$ (where $\lambda \in J$ is from [Item b\)](#)). Note that

$$\left\{ h_n(x) = \frac{f_n(x) - f_n(\lambda)}{x - \lambda} \right\}_{n=1}^{\infty} \text{ conv. uniformly on } J_{\lambda}.$$

Due to [Claim 6.7](#), since the function $g(x) = x - \lambda$ is bounded on J_{λ} ,

$$\left\{ g(x)h_n(x) = \frac{f_n(x) - f_n(\lambda)}{x - \lambda}(x - \lambda) = f_n(x) - f_n(\lambda) \right\}_{n=1}^{\infty}$$

converges uniformly on J_{λ} .

Due to [Claim 6.8](#), since

$$f_n(x) = (f_n(x) - f_n(\lambda)) + f_n(\lambda)$$

and the sequence $\{f_n(\lambda)\}_{n=1}^{\infty}$ converges uniformly on J_{λ} (since it converges due to [Item b\)](#) and does not depend on x), it follows that

$$\{f_n(x)\}_{n=1}^{\infty} \text{ conv. uniformly on } J_{\lambda}.$$

Due to [Item b\)](#), the sequence $\{f_n(x)\}_{n=1}^{\infty}$ converges pointwise on the one element set $\{\lambda\}$. Due to [Claim 6.9](#), this implies that the sequence converges uniformly on the set $\{\lambda\}$. Since the sequence converges uniformly on the sets J_{λ} and $\{\lambda\}$, due to [Claim 6.6](#) it converges uniformly on the set $J \cup \{\lambda\} = J$. Thus,

$$\{f_n(x)\}_{n=1}^{\infty} \text{ conv. uniformly on } J.$$

2) Due to [Item 1\)](#), we know that the sequence $\{f_n(x)\}_{n=1}^{\infty}$ converges uniformly on J . Let

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad (\forall x \in J). \quad (18)$$

That is, f is the limit function of the sequence $\{f_n(x)\}_{n=1}^{\infty}$ on J .

Then, for every $n \in \mathbb{N}$, consider the function

$$g_n : J_y \rightarrow \mathbb{R}, \quad g_n(x) = \frac{f_n(x) - f_n(y)}{x - y}, \quad \forall x \in J_y = J \setminus \{y\}. \quad (19)$$

Note that:

a) y is a finite limit point of J_y ;

b) $\forall n \in \mathbb{N} \rightarrow \exists \lim_{x \rightarrow y} g_n(x) = \lim_{x \rightarrow y} \frac{f_n(x) - f_n(y)}{x - y} = f'_n(y)$ (holds true due to [Item a\)](#));

c)

$$g_n(x) \xrightarrow{J_y} \frac{f(x) - f(y)}{x - y}.$$

Due to [Formula \(17\)](#), we know that the sequence $\{g_n(x)\}_{n=1}^{\infty}$ converges uniformly on J_y . The limit function can be found from [Formulas \(18\) and \(19\)](#):

$$\lim_{n \rightarrow \infty} g_n(x) = \lim_{n \rightarrow \infty} \frac{f_n(x) - f_n(y)}{x - y} = \frac{f(x) - f(y)}{x - y}. \quad (20)$$

Therefore, all conditions of [Claim 6.1](#) are satisfied and due to [Item 2\)](#) of [Claim 6.1](#), we have:

$$\lim_{n \rightarrow \infty} \left(\lim_{x \rightarrow y} g_n(x) \right) = \lim_{x \rightarrow y} \left(\lim_{n \rightarrow \infty} g_n(x) \right). \quad (21)$$

The left hand side is $\lim_{n \rightarrow \infty} f'_n(y)$ due to [\(19\)](#) and [Item a\)](#), and the right hand side is

$$\lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y}$$

due to [Formula \(20\)](#).

Due to [Item c\)](#) and [Claim 5.3](#), the limit $\lim_{n \rightarrow \infty} f'_n(y)$ exists for every $y \in J$. Therefore, due to [Equalities \(21\)](#), we have

$$\exists f'(y) \stackrel{\text{def}}{=} \lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y} = \lim_{n \rightarrow \infty} f'_n(y), \quad (\forall y \in J).$$

□

Claim 6.11 (Continuity of a Series). Let $E \subseteq \mathbb{R}$, $u_k : E \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

If:

- a) for every $k \in \mathbb{N}$, u_k is a continuous function on E ;
- b) $f_n \xrightarrow{E} f$,

then, the function f is continuous on E .

Proof. Since a finite sum of continuous functions is a continuous function (see Calculus-1),

$$f_n(x) = \sum_{k=1}^n u_k(x)$$

is continuous on E . Thus, the sequence $\{f_n\}_{n=1}^{\infty}$ satisfies all conditions of [Claim 6.2](#). Therefore, due to [Claim 6.2](#), the function f is continuous on E . □

Claim 6.12 (Term-by-term Integration of a Uniformly Conv. Series). Let $I = [a, b] \subset \mathbb{R}$, $u_k : I \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

If:

- a) for every $k \in \mathbb{N}$, u_k is Riemann integrable on I ;
- b) $f_n \xrightarrow{I} f$,

then

$$\sum_{k=1}^{\infty} \int_a^b u_k(x) dx = \int_a^b \left(\sum_{k=1}^{\infty} u_k(x) \right) dx = \int_a^b f(x) dx.$$

Proof. Since a finite sum of Riemann integrable functions is a Riemann integrable function (see Calculus-1), the function

$$f_n(x) = \sum_{k=1}^n u_k(x)$$

is Riemann integrable on I . Thus, the sequence $\{f_n(x)\}_{n=1}^\infty$ satisfies all conditions of [Claim 6.5](#). Therefore, we have:

$$\begin{aligned} \sum_{k=1}^\infty \int_a^b u_k(x) dx &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_a^b u_k(x) dx \\ &= \lim_{n \rightarrow \infty} \int_a^b \sum_{k=1}^n u_k(x) dx \\ &= \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx \\ &= \int_a^b \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx \\ &= \int_a^b \left(\lim_{n \rightarrow \infty} \sum_{k=1}^n u_k(x) \right) dx \\ &= \int_a^b f(x) dx, \end{aligned}$$

where: 1) due to [Def 1.3](#); 2) since the sum is finite; 3) and 5) since $f_n(x) = \sum_{k=1}^n u_k(x)$; 4) due to [Claim 6.5](#); 6) since $f_n \xrightarrow{I} f$ due to [Item b\)](#) and [Claim 5.3](#). $\square$

Claim 6.13 (Term-by-term Differentiation of Series). Let $J = (a, b) \subset \mathbb{R}$, $u_k : J \rightarrow \mathbb{R}$, $\forall k \in \mathbb{N}$, and

$$f_n(x) = \sum_{k=1}^n u_k(x), \quad \forall n \in \mathbb{N}.$$

If:

- a) for every $k \in \mathbb{N}$, $u_k(x)$ is differentiable on J ;
- b) there is $\lambda \in J$ such that the series $\sum_{k=1}^\infty u_k(\lambda)$ converges;
- c) the sequence $\{f'_n(x)\}_{n=1}^\infty$ converges uniformly on J ,

then:

- 1. $f_n \xrightarrow{J} f$;
- 2. the limit function f is differentiable on J and

$$\sum_{k=1}^\infty u'_k(x) = \left(\sum_{k=1}^\infty u_k(x) \right)' = f'(x), \quad (\forall x \in J).$$

Proof. Follows directly from [Claim 6.10](#). $\square$

Claim 6.14 (Stone-Weierstrass Theorem). Let $f : [a, b]$ be a continuous on $[a, b]$ function. Then, for every $\varepsilon > 0$ there is a polynomial function

$$p : [a, b] \rightarrow \mathbb{R}$$

such that

$$\forall x \in [a, b] \rightarrow |f(x) - p(x)| < \varepsilon.$$

Proof. Without proof, this is not a simple theorem.

□